AI-Native Self-Healing Open-Radio Access Network (O-RAN) Network using Digital Twin

AI-Native Self-Healing O-RAN Network using Digital Twin proposes an autonomous network management framework that combines AI-based anomaly detection with a digital twin to predict, validate, and resolve RAN faults in a closed loop. The project targets improved reliability, faster recovery, and lower operational effort for future O-RAN deployments.

Problem Statement

O-RAN networks are becoming more complex and distributed, which makes manual fault diagnosis and recovery slower and less reliable. Traditional static alarms or rule-based responses often react too late, require human intervention, and may not identify the true root cause of a problem. The project aims to solve this by combining AI and a digital twin to create a closed-loop self-healing mechanism that can sense, predict, decide, and verify recovery actions.

Worklet objective

- Detect anomalies in O-RAN KPIs using AI-based monitoring.
- Predict faults before they cause major service disruption.
- Use a digital twin to simulate network conditions and validate healing actions safely.
- Automate corrective actions such as parameter tuning, rerouting, or recovery workflows.
- Compare the AI-native approach with traditional fault management methods.

References

Research on **Closed-Loop Autonomous Network Healing Using Digital Twins**, which directly matches the project theme.
The overview of **Digital Twin for O-RAN Towards 6G**, which explains why digital twins are useful for open and autonomous RANs.
Material on **AI in O-RAN** describing predictive maintenance, anomaly detection, and self-healing capabilities.
Work on **anomaly detection in Open RAN** showing how digital twins support controlled KPI streams, repeatable fault injection, and model training.
Broader AI-native autonomy discussions for O-RAN and 6G evolution.
https://www.3gpp.org/technologies/5g-system-overview (Basic)

Students:

Tanmaya Kumar	bl.en.u4eac23058@bl.students.amrita.edu
Raghu Ram K	bl.sc.u4cse24142@bl.students.amrita.edu
Munipalle Jaswanth Kumar	bl.sc.u4cse24130@bl.students.amrita.edu

Mentors:

Bikas Singh (NovaThink Tech)

Navin Kumar (k_Navin@blr.amrita.edu)
9019930325

Core idea

Build a closed-loop system where a digital twin mirrors the live O-RAN environment, detects anomalies, predicts faults, and recommends or triggers healing actions before service impact grows. Research on O-RAN digital twins and AI-driven self-healing highlights this as a practical path toward autonomous RAN operations.

Scope of work

The work should cover anomaly detection, root-cause analysis, fault prediction, and automated recovery in an O-RAN environment. The digital twin can be used to emulate network states, inject faults, test mitigation actions safely, and train AI models using realistic KPI streams. A strong PRISM scope also includes comparison against rule-based healing and evaluation under different fault scenarios such as configuration errors, traffic overload, and node/link degradation.

Expected Outcome

- A conceptual architecture for AI-native self-healing in O-RAN.
- A digital-twin-based testbed for fault injection and recovery validation.
- AI models for anomaly detection or predictive fault identification.
- Faster fault recovery and reduced downtime compared with manual or rule-based methods.
- Evaluation results showing improvements in availability, recovery time, and operational efficiency.

M1: 0-8 days	M2, & M3 (10-22 days)	M4, M5,M6 (24-36 days)	M6 (40-50 days)
Problem study and Literature Survey Define O-RAN self-healing requirements and digital twin use cases. System Architecture design Finalize data flow between live O-RAN, AI engine, and digital twin.	M2: KPI and fault taxonomy Identify KPIs, anomaly types, and healing targets. M3: Digital twin setup Build or simulate the O-RAN twin and fault injection environment.	**M4**: AI model development Implement anomaly detection or predictive fault models. **M5:** Healing logic integration Add decision engine for automated or assisted recovery actions. **M6**:Testing and evaluation Measure detection accuracy, recovery time, and service continuity.	**M6**: Paper & IP Draft (Publication + Patent disclosure) Final documentation Prepare results, findings, and future enhancement ideas

Open Fronthaul S-Plane Timing Security

Sub-scope of the AI-Native Self-Healing O-RAN Network using a Digital Twin worklet

TEAM

Tanmaya Kumar · Raghu Ram K · Munipalle Jaswanth Kumar

MENTORS

Mr. Bikas Singh (NovaThink Tech)
Dr. Navin Kumar (Amrita Vishwa Vidyapeetham)

Amrita School of Engineering, Bengaluru · Samsung PRISM

168 live on-wire runs · 14 fault scenarios · 12 randomised replicates

REVIEW RESPONSE

Reviewer Questions and Where Each Is Answered

Questions transcribed from the reviewer's handwritten notes on the PRISM review presentation | Every new slide carries numbered references [n] listed at the end of the deck

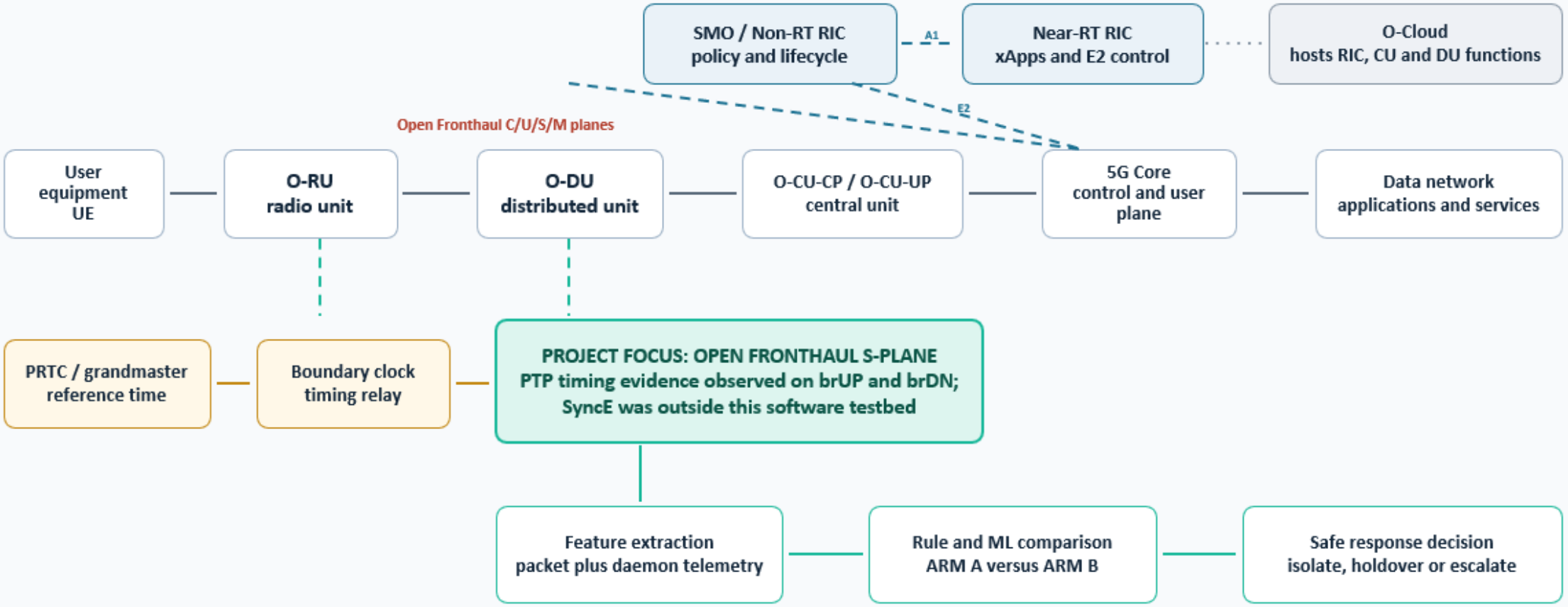
REVIEWER QUESTION	ANSWERED ON
What kind of attack, by whom, and how? Source of attack?	Threat model and entry points: slides 9–10
Were the eight attacks existing or project-defined? Citations?	Attack provenance: slide 11; references: slide 46
What are the impacts, and which module do they affect?	Affected component: slide 12; evidence boundary: slide 37
Explain the DU layers and architecture	Functional split and timing: slides 5–6
Does it apply beyond O-RAN?	Applicability: slide 44
From which model did we get the KPIs?	KPI sources: slide 20
Where did we get the dataset?	Dataset provenance: slide 23
How are KPIs classified for recovery?	KPI classes: slide 21
How does recovery work, and what did it achieve?	Recovery design/results: slides 34–35
Q16: Is the system reactive, preventive or predictive?	Mechanism classification: slide 36

AUDIT STATUS November citation errata applied; recovery and impact evidence added with source-level traceability.

SYSTEM ARCHITECTURE AND PROJECT BOUNDARY

End-to-End O-RAN Architecture and S-Plane Project Boundary

O-RAN Alliance architecture description (ETSI TS 103 982 PAS) | O-RAN.WG4.CUS.0 Control, User and Synchronization Plane Specification | 3GPP TS 38.401 NG-RAN architecture



Validated here: live packet-path execution, extracted timing evidence and classification. Remaining scope: digital-twin orchestration, fault prediction, automated recovery and outcome measurement.

SYSTEM ARCHITECTURE · PROTOCOL LAYERS

O-RAN Functional Split: What the O-DU and O-RU Host

ETSI TS 103 982 V8.0.0 (2024-01), O-RAN Architecture Description, clauses 3.1, 6.3.3–6.3.6, 6.4.7 [4] | 3GPP TS 38.401 NG-RAN architecture [11]

O-CU-CP

central unit · control plane

- RRC
- PDCP (control plane)

E1

O-CU-UP

central unit · user plane

- SDAP
- PDCP (user plane)

F1

O-DU

distributed unit

- RLC
- MAC
- High-PHY

Open FH 7-2x

O-RU

radio unit · physical node

- Low-PHY (FFT / iFFT, PRACH extraction)
- RF processing

E2 terminates at O-CU-CP, O-CU-UP and O-DU, not at the O-RU (cl. 6.3.3–6.3.6).

OPEN FRONTHAUL PLANES (TS 103 982 cl. 6.4.7)

- C-Plane · Control
- U-Plane · User
- **S-Plane · Synchronization — this project**
- M-Plane · Management

WHY THE O-RU

Radio requirements are measured at the base-station antenna [8][9]; in O-RAN the antenna side is the O-RU (Low-PHY + RF). The S-plane is how the O-RU receives time, so it is where this project works.

SYSTEM ARCHITECTURE · TIMING DISTRIBUTION

How the O-RU Receives Time: LLS-C1 to LLS-C4

O-RAN.WG4.CUS.0 v06.00 synchronization configurations, as quoted in G. Armstrong, "O-RAN Fundamentals", ATIS WSTS 2023 [7] | Exposure column reasoned from [1], [2], [10], [12]

CONFIG	DEFINITION (O-RAN.WG4.CUS.0, AS QUOTED IN [7])	WHERE AN ATTACKER CAN ACT
LLS-C1	O-DU is part of the synchronization chain. Network timing is distributed from O-DU to O-RU via a direct connection between O-DU site and O-RU site.	A compromised O-DU (acting as synchronization master in LLS-C1) or an attacker on the O-DU–O-RU link can control or disrupt the O-RU's time (threat analysis from LLS-C1 topology; cf. T-FRHAUL-02 and T-SPLANE-01…04 [2], RFC 7384 §3.2 [1]).
LLS-C2	O-DU is part of the synchronization chain. Timing goes from O-DU to O-RU; one or more Ethernet switches are allowed in the fronthaul network.	As C1, plus any device on the switched fronthaul can inject or intercept PTP [10, sec. 4].
LLS-C3	O-DU is not part of the synchronization chain. Timing is distributed from PRTC/T-GM to O-RU, typically from central or aggregation sites.	Grandmaster, boundary clocks and switches on the path are all exposed. TIMESAFE's production test used LLS-C3 [10].
LLS-C4	Synchronization reference provided to the O-RU with no involvement of the transport network, typically a local GNSS receiver.	No PTP exposure; GNSS jamming or spoofing instead (RFC 7384 sec. 3.2.10 [1]; BSI [12]).

OUR TESTBED

Grandmaster → boundary clock → three radio units over bridged segments, with no O-DU in the timing chain: the LLS-C3 structure. The O-DU itself is not emulated.

PROBLEM STUDY

Open Fronthaul Timing Requirements and Security Exposure

O-RAN.WG4.CUS.0 timing categories (via ATIS/WSTS O-RAN Fundamentals tutorial, Armstrong 2022) | ITU-T G.8271.1 network time-error budget | Groen et al., TIMESAFE, ACM ToPS 28(5), 2025, doi:10.1145/3775060

130 ns

Relative time alignment error between radio units in a cluster

Timing Category A; implies ≈±65 ns per O-RU against a common reference under equal allocation

O-RAN.WG4.CUS.0

±1.5 µs

Absolute time error at reference point E, common time standard

Total budget at reference point E; the network limit at reference point C is 1100 ns

ITU-T G.8271.1

≈2 s

Observed time from attack onset to O-RU crash in one published study

The O-RU crashed and required a manual reboot, with switch PTP ports in dynamic role

TIMESAFE, ACM ToPS 2025

WHY THIS IS A SECURITY SURFACE, NOT ONLY AN ENGINEERING CONSTRAINT

Time is distributed across the fronthaul as ordinary Ethernet frames carrying the IEEE 1588 protocol, not over a dedicated physical channel. Any device able to place frames on that path can therefore advertise itself as a time source, forge timing content, or withhold legitimate messages. Loss of a shared time reference degrades radio coordination, so the S-plane is simultaneously a timing-performance boundary and an attack

Figure basis — **The 130 ns and ±1.5 µs values derive from 3GPP air-interface requirements re-expressed by O-RAN and ITU-T. The ≈2 s figure is a single observed experimental result, not a standardised limit.**

DECISION REQUIREMENT

Attack–Fault Ambiguity and Sub-Scope Selection

IEEE 1588-2019 clock states and timePropertiesDS | ITU-T G.8275.1 clockClass ladder (6 locked, 7 holdover in spec) | O-RAN WG11 Threat Model, ETSI TR 104 106 V3.0.0

BENIGN FAULT

The clock is wrong

Example: a GNSS outage at the grandmaster, declared through clockClass 6 → 7.

Correct response — ride it out in holdover.

Isolating the node here creates an outage that was not going to occur.

ATTACK

The clock is wrong

Example: a rogue master injecting false time through forged Announce messages.

Correct response — isolate the source immediately.

Waiting in holdover allows the attacker to keep steering the clock.

Identical observable symptom · opposite correct response · the decision cannot be made from the symptom alone

RATIONALE FOR SELECTING THE S-PLANE SUB-SCOPE

01 Highest consequence

A timing failure can become a service failure within seconds, leaving little time for a safe decision.

02 Genuine ambiguity

Selected benign faults and attacks produce overlapping symptoms; the decision needs provisioned context.

03 Constrained by standards

IEEE 1588-2019 and ITU-T G.8275.1 fix wire-level legality, giving objective decision criteria.

04 Reproducible in software

Real PTP daemons can be executed and attacked on a virtual wire without a proprietary RAN stack.

The sub-scope was selected because it is the point where a detection label alone is insufficient: the same symptom requires opposite recovery actions, and standards supply objective criteria for separating them.

Who Attacks the S-Plane: Agents and Classes

ETSI TR 104 106 V3.0.0 (2025-06) clause 7.2 threat agents and clause 7.4.1.2 [2] | IETF RFC 7384 (2014) clause 3.1 threat model [1] | TIMESAFE, ACM TOPS 2025, section 4 [10]

THREAT AGENTS (ETSI TR 104 106, CL. 7.2)

Insiders	malicious attack by a person with authorized access
Nation-state	aggressively target and gain persistent access to networks
Cyber-criminals	individuals who commit cybercrimes
Hacktivists	cyber-attacks for political or social gains
Cyber-terrorists	target O-RAN infrastructure through violence
Script kiddies	actors without deep technical expertise

ATTACKER CLASSES BY ACCESS AND POSITION (RFC 7384, CL. 3.1)

Internal	has access to a trusted segment of the network, or holds the keys
External	no keys; sees only encrypted or authenticated traffic
Man-in-the-middle	positioned to intercept and modify in-flight packets
Injector	not in-path; attacks by generating protocol packets

HOW THEY GET ON THE FRONTHAUL (threat-model synthesis from [10] §4 and [2] T-FRHAUL-02)

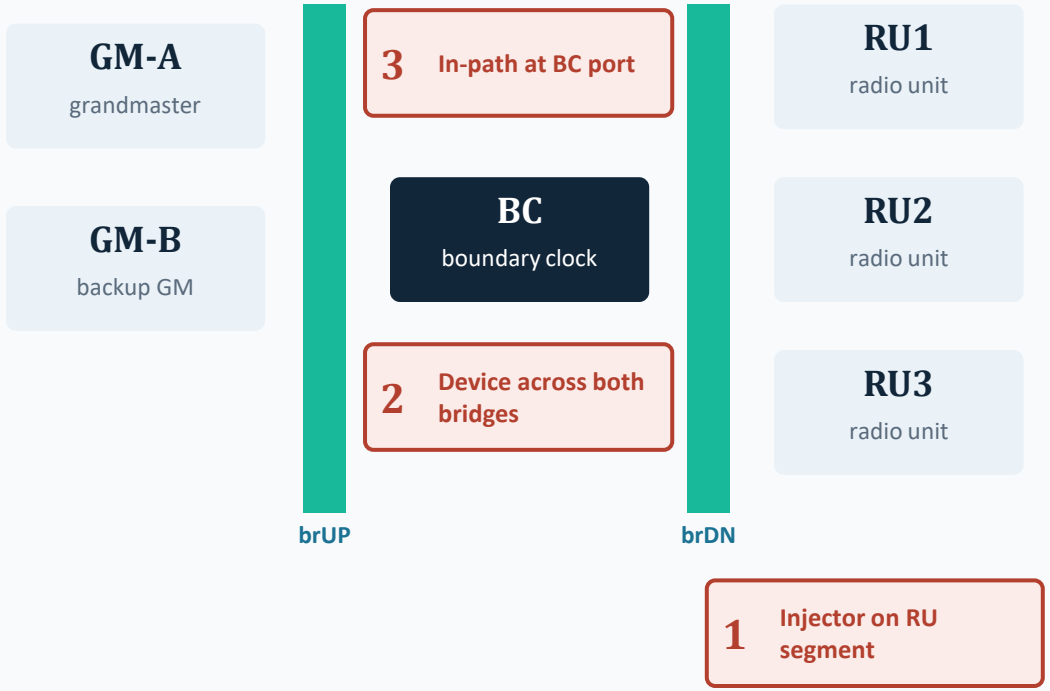
- Remote access widened by multi-vendor, disaggregated equipment ([10] §4)
- Physical access to edge cell sites hosting O-RU / O-DU ([10] §4)
- Dependency and supply-chain vulnerabilities in DU or RU equipment ([10] §4, citing [37, 55])
- Unauthorized access to the Open Fronthaul Ethernet L1 physical interface ([2] T-FRHAUL-02)

ANSWER: WHO

The standards name attackers by capability and position, not by identity. Six of our eight attacks fall in RFC 7384 rows that require an internal attacker; interception (C1) can also be done by an external man-in-the-middle.

Attack Entry Points on the Testbed Timing Path

Positions and tools read from the campaign scripts run/scenarios.sh and run/run_gap.sh in splane_campaign_CORRECTED_2026-09-20.tgz | Attacker classes per RFC 7384 cl. 3.1 [1]



#	SCENARIOS	HOW IT WAS LAUNCHED
1	A1, A2, A3, A5, C2, C3	Device on the RU-side fronthaul segment (the ru3 namespace, or an added namespace on brDN). A1: second ptp4l with off-allow-list identity; A2, A5, C2, C3: Scapy-built frames; A3: tcpdump capture, then tcpreplay.
2	A8	Two-port ptp4l bridging brUP and brDN that slaves to the real GM and re-advertises downstream with stepsRemoved incremented.
3	C1	Boundary clock's downstream port set down for 60–75 % of the injection window (as drawn): every frame is removed, not just selected message types.

RFC 7384 position: 1 = internal injector · 2 = internal man-in-the-middle (rogue master) · 3 = man-in-the-middle (internal or external)

SCOPE

All attackers are Linux network namespaces on one host; no real network or host clock was touched. Real-world access paths are those on the threat-model slide.

Where Each of the Eight Attacks Comes From

IETF RFC 7384 cl. 3.2 [1] | ETSI TR 104 106 V3.0.0 T-SPLANE-01…04 [2] | ETSI TS 104 105 V7.0.0 O-RAN Security Test Specifications [3] | TIMESAFE sec. 4.1 [10]

ID	ATTACK	PUBLISHED THREAT	PUBLISHED TEST OR STUDY	WHAT WE CHOSE
A1	Rogue grandmaster	RFC 7384 §3.2.4 rogue master; T-SPLANE-03	TS 104 105 §11.1.5.2.2 Rogue PTP Instance; TIMESAFE spoofing	Randomised identity, priority2, clockClass
A2	Sync / Follow_Up spoofing	RFC 7384 §3.2.2 spoofing	No ETSI test of this exact form	Off-allow-list source, bursts, no Announce
A3	Replay	RFC 7384 §3.2.3 replay	TIMESAFE replay attack	Live capture replayed with tcpreplay
A5	DoS / PTP flooding	RFC 7384 §3.2.9 time-protocol DoS; T-SPLANE-01	TS 104 105 §11.1.5.1.1, §24.2.1.1	Burst size and gap randomised
A8	Rogue boundary clock	RFC 7384 §3.2.4 rogue master, in MITM position	Adapted: no TS 104 105 test places the rogue as a relay	Relays the real GM, stepsRemoved +1
C1	Interception and removal	RFC 7384 §3.2.5; T-SPLANE-04	TS 104 105 §11.1.5.3.1	Realised as a port blackhole (all frames)
C2	Malformed frames	RFC 7384 §3.2.11 exploiting vulnerabilities	Category related to TS 104 105 §24.2.1.2 S-Plane PTP Unexpected Input (O-DU 250,000-iteration fuzz test)	Project-defined 3-variant deterministic subset (versionPTP 3, reserved type, short length; not a reproduction of §24.2.1.2)
C3	Whole-second field abuse	RFC 7384 §3.2.2 spoofing (false time)	None found; fields per IEEE 1588-2019 §8.2.4	Our construction: leap61 set, UTC offset 0

ANSWER

None of the eight threat classes was invented. Seven follow a published threat or test; C3 is our own instance of the RFC 7384 spoofing class, with no published test of that exact form.

THREAT MODEL · IMPACT

Which Component Each Attack Affects, and What

Measured: ptp4l logs of RU1–RU3, 12 replicates per scenario (36 RU logs each), extracted_168run_ptp4l_logs | Impact classes: RFC 7384 Table 1 [1]; ETSI T-SPLANE impacts [2]; TIMESAFE [10]

ID	AFFECTED COMPONENT	IMPACT CLASS	MEASURED IN OUR RUNS	REPORTED IN LITERATURE
A1	All RU slave clocks on the segment	False time	36/36 RU logs selected the rogue as best master	TIMESAFE: O-RU crash ≈2 s after spoofing, cell drop, manual reboot
A8	All RUs behind the inserted clock	False time	36/36 RU logs recorded the rogue BC as a foreign master; parent unchanged (5/5 pmc runs, 5 Oct)	T-SPLANE-03: slaves given inaccurate time
C1	RUs downstream of the BC port	Accuracy loss, DoS	36/36 RU ports lost master on Announce timeout; servo reports 7.8 vs 22 per log	T-SPLANE-04: degradation or free-running
C3	RU parent and time properties	False time	RUs re-parented to the forger under GM-A's identity (5/5 pmc runs, 5 Oct); servo reports 3.0 vs 22 per log	Spoofing class: false time
A2 A3 A5 C2	RU receive path	False time / DoS	No master change in any RU log or pmc run; visible only in packet evidence	A5: T-SPLANE-01 clock service interrupted; A3: TIMESAFE replay

Service limits at stake: TDD cell phase sync better than 3 µs (3GPP TS 38.133 [8]) · TAE ≤ 65 ns for MIMO, ≤ 260 ns intra-band contiguous CA (3GPP TS 38.104 §9.6.3.2 [9])

LIMIT

Every daemon ran with free_running 1: offsets were measured, never steered. Measured impact is protocol-level; time-error and service impact come from the cited literature.

LITERATURE SURVEY

Research Foundation for S-Plane Timing Assurance

Groen et al., TIMESAFE, ACM ToPS 28(5), 2025, doi:10.1145/3775060 | Bonati et al., Colosseum, IEEE OJ-COMS, 2024, doi:10.1109/OJCOMS.2024.3447472
Digital Twin for O-RAN, IEEE Communications Magazine, 2024, doi:10.1109/MCOM.003.2400016 | OpenRAN Gym, Computer Networks, 2023 | TSN Digital Twin, EURASIP JIS, 2025, doi:10.1186/s13635-025-00213-7

SOURCE	WHAT THE LITERATURE ESTABLISHES	BOUNDARY LEFT OPEN
TIMESAFE · ACM ToPS 2025	PTP attacks disrupt O-RAN fronthaul and are detectable from packet evidence.	Detection was not tested against a matched set of benign timing faults.
Digital Twin for O-RAN · IEEE 2024	A RAN twin supports testing, optimization and safer control decisions.	The architecture does not validate S-plane attack-versus-fault discrimination.
Colosseum · IEEE 2024	Repeatable Open RAN twins reproduce realistic experimental conditions.	The platform does not address telecom PTP legality or recovery choice.
OpenRAN Gym · Elsevier 2023	O-RAN AI/ML requires controlled data collection and reproducible evaluation.	It does not test whether an S-plane model beats rules or a constant predictor.
TSN Digital Twin · Springer 2025	A timing-aware twin supports structured security testing.	The study does not combine O-RAN PTP, benign lookalikes and held-out controls.

Prior work establishes the threat, the AI testbed direction and the digital-twin direction. It does not resolve the operational decision between malicious timing behaviour and a benign timing fault presenting the same symptom.

RESEARCH GAP ANALYSIS

Research Gaps Behind the Experimental Design

Federated Continual Learning for O-RAN Anomaly Detection, IEEE WCNC 2024 | Adversarial ML Threat Analysis in O-RAN, JNCA 2024, doi:10.1016/j.jnca.2024.104090
O-RAN xApps: Survey and Research Challenges, Computer Networks 2025 | Anomaly Detection for xApp and E2 Threats, IEEE 2025

01

Attack and fault ambiguity

Prior detectors separate attack from normal operation. Operators must also separate attacks from real clock, path and configuration faults.

02

Unverified AI claims

Reported model accuracy can appear strong without a held-out split, a rule comparator and a trivial constant baseline.

03

Protocol evidence separated from ML

Standards checks and statistical models are usually presented as alternatives rather than evaluated on the same executed runs.

04

Detection separated from safe response

A label alone does not determine whether to isolate a source, enter holdover or escalate to an operator.

RESULTING STUDY DESIGN

Matched scenarios

Attacks and benign lookalikes execute through the same packet path.

Two decision arms

ARM A applies standards legality plus operator context. ARM B applies Random Forest with an Isolation Forest novelty layer.

Held-out evaluation

Replicates 9–12 remain outside training for the 56-run final test.

Trivial control

An always-BENIGN predictor establishes whether ARM B adds information.

Explicit abstention

Ambiguous evidence escalates instead of forcing an unsafe answer.

Gap addressed: the reviewed source set did not combine S-plane benign lookalikes, standards-aware rules, a machine-learning arm, explicit abstention and a constant baseline within one held-out campaign.

CONTRIBUTION POSITIONING

Standards-Grounded Positioning of the Contribution

Normative basis: IEEE 1588-2019, ITU-T G.8275.1, O-RAN.WG4.CUS.0, O-RAN WG11 (ETSI TR 104 106 V3.0.0) | Prior-art boundary: TIMESAFE, Colosseum, OpenRAN Gym, digital-twin literature

01
Normative ground

Protocol legality and telecom timing expectations are taken from recognized standards, not from arbitrary thresholds.

02
Executed evidence

Real PTP daemons, packet captures and daemon telemetry produce auditable observations rather than synthetic rows.

03
Controlled comparison

Rules, machine learning and an always-BENIGN control are scored on the same 56 held-out runs.

04
Governed response

ATTACK, BENIGN and UNKNOWN map to isolation, holdover and escalation policy respectively.

DEMONSTRATED IN THIS CAMPAIGN

Live PTP packet-path execution
Held-out attack and benign-fault classification
ARM A versus ARM B versus constant control
Explicit abstention with evidence provenance

NOT YET DEMONSTRATED

Digital-twin action verification
Executed isolate or holdover recovery
Measured MTTR and availability improvement
An ML model exceeding transparent baselines

Contribution: a standards-aware, controlled S-plane benchmark that tests whether a learned model adds decision value before it is granted authority over recovery.

STANDARDS FOLLOWED

Telecom Standards and Profile Grounding

Normative basis: IEEE Std 1588-2019 | O-RAN.WG4.CUS.0 | O-RAN WG11 Security Threat Modeling (ETSI TR 104 106 V3.0.0) | ITU-T G.8275.1
Timing performance context: ITU-T G.8273.2 | Frequency synchronization context: ITU-T G.8262 (SyncE, not implemented here)

AUTHORITY	WHAT IT DEFINES	HOW THIS PROJECT USES IT
IEEE 1588-2019	Legal PTP message fields, data sets, clock behaviour and leap-second handling.	ARM A checks field legality, sequence continuity and timePropertiesDS consistency.
O-RAN.WG4.CUS.0	Open Fronthaul C/U/S-plane architecture and synchronization configurations.	Defines the O-DU–O-RU S-plane boundary used by the experiment.
O-RAN WG11	Open Fronthaul threat model, including malicious timing-source behaviour.	Supplies the threat identifiers that label the attack scenarios.
ITU-T G.8275.1	Telecom PTP profile for phase and time distribution with full timing support.	Fixes domain, message rates, transport and the alternate BMCA used by the testbed.
ITU-T G.8273.2	Timing characteristics and limits for telecom boundary clocks.	Provides performance context for interpreting timing-quality evidence.

Scope disclosure

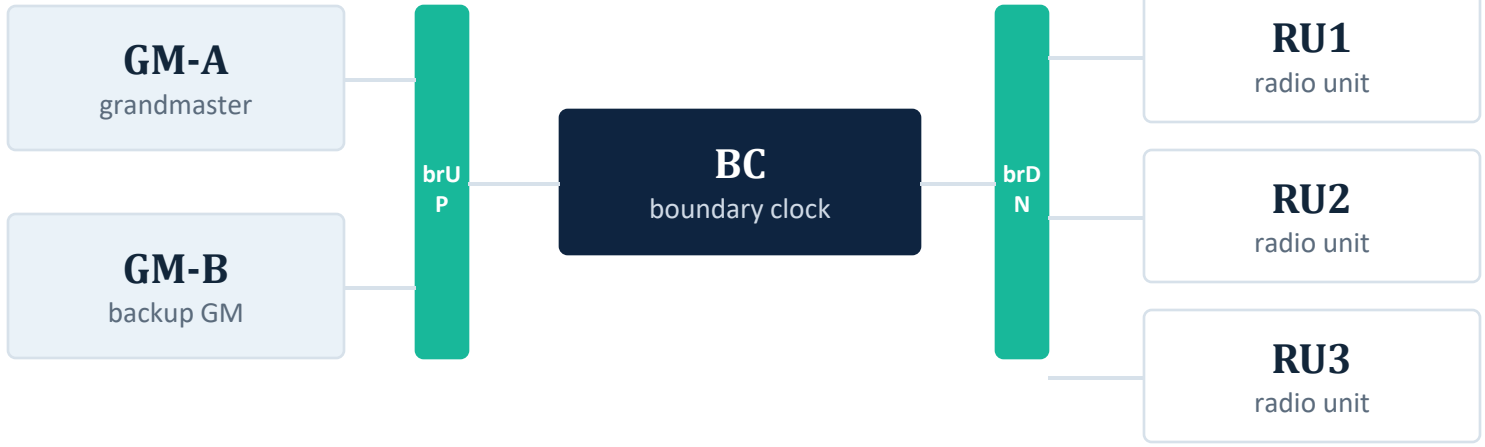
The testbed exercised PTP only; SyncE is a physical-layer function and was not implemented. ITU-T recommendations are paywalled, so several values were confirmed against authoritative secondary sources, as recorded in the project workbooks.

EXPERIMENTAL CONFIGURATION

Open Fronthaul Timing Testbed Architecture

linuxptp v4.0 (released 2023-06-09), reference implementation of IEEE 1588-2019 and ITU-T G.8275.1 | Linux network namespaces, veth pairs and software bridges | tcpdump packet capture

Six genuine linuxptp 4.0 daemons in isolated Linux network namespaces across two bridged segments. All protocol behaviour — BMCA election, servo control, message cadence — is executed by real daemons, not simulated.



CAPTURE AND TELEMETRY

tcpdump on both bridges during each run (raw PCAPs not kept in the packaged archive)
ptp4l servo logs from all six daemons
pmc management queries per run (returned no data; see KPI sources)
context.json — ground-truth label plus provisioned operator context

ITU-T G.8275.1 profile as configured

domain 24 · priority1 128 · Sync 16/s (logInterval −4) · Announce 8/s (logInterval −3)

Evidence boundary

No independent clock-error instrument was used; synchronization impact is inferred from protocol and servo telemetry, not measured against a reference clock.

Faults are produced by live packet-path mechanisms — competing daemons, injected frames, replayed captures and link or queue manipulation — never by editing rows in a dataset.

CONFIGURATION AND PLATFORM LIMITS

Timing Configuration and Verified Platform Constraints

ITU-T G.8275.1 telecom profile (values cross-checked against Cisco and Nokia profile documentation and linuxptp configs/G.8275.1.cfg) | Constraints confirmed by direct query of the running system

PARAMETER	CONFIGURED VALUE	BASIS
Profile	ITU-T G.8275.1 telecom profile	Full timing support from the network
domainNumber	24	G.8275.1 permitted range 24–43
priority1	128 (fixed)	G.8275.1 fixes the value and the alternate BMCA excludes it from comparison
Sync rate	16 / s (logInterval –4)	G.8275.1
Announce rate	8 / s (logInterval –3)	G.8275.1
BMCA	dataset_comparison G.8275.x	Alternate BMCA, not the IEEE 1588 default
Transport	L2 multicast 01-1B-19-00-00-00	Forwardable address, permitted by G.8275.1 (linuxptp default)
Timestamping	software	Platform limit — see constraints

VERIFIED PLATFORM CONSTRAINTS

! **No PTP hardware clock**
Is /dev/ptp* returns empty. Software timestamping only: a microsecond-scale noise floor against a nanosecond-scale target.

! **No loadable kernel modules**
modprobe absent and sch_netem unavailable, so congestion uses tbf.

! **One shared oscillator**
All namespaces share the host clock, and free_running 1 is set: offset is measured, never steered.

! **No GNSS receiver**
GNSS spoofing and jamming cannot be produced at all.

Effect on scope — **These constraints set the ceiling on every result that follows: five of the sixteen catalogued fault classes cannot be produced here, and no claim is made about absolute time-error magnitude.**

Every configured value was read back from the running system and traced to the governing standard or, where the normative text is paywalled, to an authoritative secondary source.

PROCESSING PIPELINE

Data Flow from Packet Capture to Verdict

Deep extractor: 56 fields per PTP frame | Field definitions from IEEE 1588-2019 message and data-set clauses | Conformance flags derived from ITU-T G.8275.1 profile constraints

01

The wire

Real PTP frames produced by six linuxptp daemons on two bridged segments

02

Capture

tcpdump captures PTP frames on brUP and brDN; raw PCAPs not kept in the archive

03

Deep extract

56 protocol and conformance fields per packet; fields that cannot be observed are written EMPTY

04

Decision rule

Standards legality plus provisioned operator context, applied as ordered clauses

05

Verdict

ATTACK · BENIGN · UNKNOWN, with the triggering clause recorded

INTEGRITY RULE CARRIED THROUGH THE WHOLE PIPELINE

A value that cannot be measured is written EMPTY — never zero.

A zero offset reads as perfect synchronization. An earlier build wrote zero for unobserved fields and consequently reported a blind node as healthy. Nothing is treated as healthy without positive proof of observation, and this rule is applied at extraction, at feature construction and at scoring.

The same extracted evidence feeds both decision arms, so the rule-based and machine-learning comparison is performed on identical executed runs rather than on separately prepared datasets.

KPI SOURCES

KPIs Are Measured From the Protocol, Not a Model

IEEE 1588-2019 message formats and data sets [5] | linuxptp ptp4l and pmc [14] | Extractor ptp_deep_extract.py; logs extracted_168run_ptp4l_logs; pmc_log.sh

01

Wire capture

tcpdump on brUP / brDN → ptp_deep_extract.py

56 fields per PTP message: header, flags, correctionField, sourcePortIdentity, sequenceId, Announce data set, G.8275.1 conformance flags

Used by the rule (ARM A)

02

Daemon servo log

ptp4l -m output on each node

master offset (ns), servo state, frequency adjustment (ppb), path delay (ns), port-state transitions

Used by the ML model (ARM B)

03

Management queries

pmc GET on 7 data sets per node, every second

Intended: CURRENT, PARENT, TIME_PROPERTIES, PORT data sets and port statistics

Returned no data in any run

04

Operator context

context.json written per run

Provisioned allow-list, expected boundary clock, maintenance-window state, ground-truth label

Configuration, not measurement

Example servo line, baseline run 1, RU1: `ptp4l[517.414]: master offset 1191 s0 freq -1547 path delay 5313`

ANSWER

No model generates the KPIs. They are read from PTP itself, defined by IEEE 1588-2019 and ITU-T G.8275.1. The ML model consumes six servo-derived features; it does not produce KPIs.

KPI CLASSIFICATION FOR RECOVERY

KPI Classes and the Recovery Decision Each One Drives

Classes derived from the clause order of run/decision_rule.py (base) and run/decision_rule_v3.py | Field legality per IEEE 1588-2019 [5] and ITU-T G.8275.1 [6] | Response labels as defined on the classification-method slide

KPI CLASS	KPIs (FIELD, LOG OR CONTEXT-KEY NAMES)	WHAT IT REVEALS	RECOVERY DECISION
Protocol legality	version_ptp, message_type_raw, message_length, clock_class_legal, g87251_domain_ok, g87251_mcast_ok, g87251_priority1_ok, steps_removed ≥ 255, flag_alternateMaster	Structurally illegal traffic (C2; illegal Announce)	ATTACK → ISOLATE source
Identity and topology	source_clock_identity, gm_clock_identity, gm_priority2, gm_clock_class, steps_removed vs provisioned inventory	Rogue GM (A1), rogue BC (A8), unknown sender (A2)	ATTACK → ISOLATE source
Continuity and rate	sequence_id regressions; delivered message rate vs the rate declared in log_message_interval	Replay (A3), flooding (A5), removal (C1)	ATTACK → ISOLATE source
Time properties	flag_leap61 / leap59, current_utc_offset, flag_currentUtcOffsetValid, flag_timeTraceable	Whole-second abuse (C3)	ATTACK → ISOLATE source
Operator context	context.json: gm_allowlist, expected_bc_identity, maintenance_window_open	Planned vs unplanned change (B2, B7, B_unplanned)	BENIGN → TOLERATE / HOLDOVER · UNKNOWN → ESCALATE
Timing quality	ptp4l servo log: master offset, path delay; derived offset_std, pdv_std	Degradation magnitude; not used for the verdict	Severity input only
Not observable here	GNSS status, SyncE quality level, oscillator error	Needed for A6, A7, B1, B4, B6	Requires hardware

READING ORDER

Legality, identity, continuity and time-property KPIs decide ATTACK; context separates planned from unplanned changes; timing quality grades severity. Actions are recommended, not executed, in the campaign.

DATASET PRODUCED

Experimental Dataset Composition

Packaged archive: splane_campaign_CORRECTED_2026-09-20.tgz, 1096 files, sha256 6149b4fb15940cdac694ba8666d97041b4eb62d5523d2631c1edf96d531e4ed9 | Run inventory enumerated in EVALUATION_V4.json

168	14	12	1,451,909	56
live on-wire runs	fault scenarios	randomised replicates each	PTP packet records extracted	fields per packet

WHAT CONSTITUTES ONE RUN

- **Two bridge captures** — brUP and brDN; raw PCAPs not kept in archive
- **One deep-extract CSV** — 56 fields per packet; unobservable fields written EMPTY
- **Six daemon servo logs** — offset, path delay and frequency per node
- **One pmc telemetry file** — GET requests logged; no responses returned in any run
- **One context.json** — ground-truth class, fault id, allow-list, maintenance state
- **Three verdict files** — base rule, v2 and v3 scored on the same capture

MEASURABLE AND UNAVAILABLE EVIDENCE

Recorded for every run

Packet header, flag, correction, identity, sequence and Announce data-set fields; derived G.8275.1 conformance flags; per-node servo telemetry.

Structurally unavailable

GNSS status, SyncE quality level, oscillator frequency error and absolute clock error. No hardware exists in this testbed to produce them, so the corresponding columns are EMPTY rather than defaulted.

Configured, returned nothing

pmc queries returned no data in any of the 168 runs.

RECORD COUNT VARIES WITH THE FAULT, AND THE VARIATION IS ITSELF EVIDENCE

C1 interception averages 2,360 records per run because the attack starves the link · baseline averages 6,021 · A5 flooding averages 38,319 because the attack adds traffic. Record volume is therefore a discriminating feature, not a data-quality artefact.

Archive contents — **Roughly 222 MB of source PCAPs were captured across the campaign. The packaged review archive retains the extracted CSVs, verdicts, context files, daemon logs, scripts and freeze records; it does not retain the source PCAPs.**

DATASET PROVENANCE

Where Every Dataset Came From and How It Is Used

dataset/README.md (classification of 16 Sep) | TIMESAFE data repository github.com/genesys-neu/s-plane_security [10] | outputs/empirical_software_network_pilot_v1/START_HERE_FINAL.md

DATASET	ORIGIN	USED FOR	STATUS
168-run campaign	Generated by this team on the software testbed (six linuxptp 4.0 daemons, G.8275.1), 20 Sep 2026. Archive sha256 6149b4fb…, 1,096 files, 1,451,909 packet records.	Every result in this deck	PRIMARY EVIDENCE
Testbed configs	cfg/*.cfg for all nodes in g87251_testbed_v2.tgz (17 Sep); not inside the corrected campaign archive.	Configuration provenance	REFERENCE
TIMESAFE captures	Public data of Groen et al., real hardware: Foxconn O-RU, NVIDIA Aerial DU, Dell S5248F-ON switch, Qulsar QG-2 grandmaster [10].	August model work; profile cross-check	REFERENCE ONLY
13 Sep pilot	Second software testbed: 25 S14 runs, 232,403 packets; S11 closed-loop trial; S15 validation not established.	Closed-loop feasibility	DEVELOPMENT DATA
Simulator output	Synthetic physics-simulator rows; quarantined in dataset/illegitimate.	Software testing only	NOT EVIDENCE

ANSWER

The evaluated dataset was generated by us on a reproducible testbed. The only external data, TIMESAFE, is public and cited; August results that used it were withdrawn on 17 Sep.

FAULT INJECTION

Fault Injection Mechanisms on the Live Packet Path

IEEE 1588-2019: message formats, sequenceId continuity, timePropertiesDS §8.2.4, leap-second handling §9.4 | O-RAN WG11 security test identifiers (ETSI TS 104 105) | Scapy, tcpreplay, Linux tc

Additional real PTP daemons

A1 · A8

A new namespace runs a genuine ptp4l instance with an identity outside the provisioned allow-list and a superior data set, then competes for election under the G.8275.1 alternate BMCA. The attack is won on protocol merit, not by editing a capture.

Crafted frame injection

A2 · C2 · C3

Hand-built frames placed on the RU segment: forged two-step timing messages; malformed frames using versionPTP 3, a reserved message type and a short length; Announce frames with leap61 set and the UTC offset falsified.

Capture and replay

A3

tcpdump captures live frames mid-run and tcpreplay retransmits them, producing genuinely stale origin timestamps and reused sequence identifiers rather than synthetic duplicates.

Link and queue manipulation

A5 · C1 · B3 · B7

High-rate flooding; sustained blackholing of the boundary clock's downstream port; a tbf bottleneck carrying competing non-PTP traffic; and bouncing an RU link during a maintenance window.

An attack and its benign twin frequently share the same packet-path mechanism. Killing a grandmaster is identical in B2 and B_unplanned_failover; only the provisioned maintenance context differs. The context is therefore part of the evidence, not metadata.

Attack Scenarios: Identity and Message Manipulation

IEEE 1588-2019 Announce, Sync and Follow_Up message clauses and sequenceId continuity | ITU-T G.8275.1 alternate BMCA and message-rate cadence | O-RAN WG11 threat identifiers, ETSI TR 104 106 V3.0.0

SCENARIO	INJECTED CONDITION	OBSERVABLE EFFECT	EXPECTED	RESULT
A1 Rogue grandmaster / BMCA spoof **T-SPLANE-02 / -03**	A second real ptp4l instance with an identity outside the provisioned allow-list and a superior data set (randomised priority2 and clockClass) competes in the G.8275.1 alternate BMCA.	The rogue clock wins the alternate BMCA and becomes the advertised time source for the downstream segment.	ATTACK	12/12 correct
A2 Sync / Follow_Up spoofing **IEEE 1588-2019**	Forged timing messages from an unallow-listed source identity, sent in bursts, with implausible origin timestamps and no accompanying Announce.	Timing content conflicts with the elected grandmaster while the source never advertises itself legitimately.	ATTACK	12/12 correct
A3 Replay **IEEE 1588-2019 sequenceId**	Live frames captured mid-run and retransmitted with tcpreplay, carrying genuinely stale origin timestamps and reused sequence identifiers.	Sequence continuity regresses and timestamps move backwards relative to the active stream.	ATTACK	12/12 correct
A5 DoS / PTP flooding **T-SPLANE-01**	High-rate PTP traffic injected on the RU segment in bursts of 300 frames, from a source outside the allow-list.	Message rate far exceeds the profile cadence; record volume rises to roughly 38,000 per run against a baseline of 6,000.	ATTACK	12/12 correct

All four scenarios were detected by the pre-frozen base rule at 12/12, before any rule revision. Detection rests on protocol legality and allow-list membership, not on the magnitude of the timing error produced.

Attack Scenarios: Relay, Interception and Field Abuse

O-RAN WG11 security test identifiers 11.1.5.3.1 and 24.2.1.2 (ETSI TS 104 105) | IEEE 1588-2019 stepsRemoved, timePropertiesDS §8.2.4 and leap-second handling §9.4 | ETSI TR 104 106 V3.0.0

SCENARIO	INJECTED CONDITION	OBSERVABLE EFFECT	EXPECTED	RESULT
A8 Rogue boundary clock T-SPLANE-03	A two-port ptp4l instance slaves to the legitimate grandmaster upstream and re-advertises downstream with stepsRemoved incremented.	An unauthorised clock is inserted into the timing path while still relaying genuine time, so content alone appears valid.	ATTACK	12/12 correct
C1 Selective interception and removal T-SPLANE-04	The boundary clock's downstream port is blackholed for a sustained fraction of the observation window, withholding Sync and Announce.	Radio units receive messages below the rate the boundary clock itself declares; delivered-to-declared ratio falls toward 0.37.	ATTACK	11/12 correct
C2 Malformed frames (project 3-variant subset; cf. WG11 24.2.1.2 unexpected-input category)	Three deterministic project-constructed malformed variants injected with versionPTP 3, a reserved message type, and a message length shorter than the protocol permits (distinct from ETSI TS 104 105 §24.2.1.2's 250,000-iteration random-mutation O-DU fuzz test).	Frames are structurally illegal under IEEE 1588-2019 regardless of their timing content.	ATTACK	12/12 correct
C3 Whole-second field abuse IEEE 1588-2019 §8.2.4	Announce frames carrying leap61 asserted, currentUtcOffset falsified to 0 against a true value of 37, and traceability flags stripped.	Measured on testbed: Announce time-properties fields become internally inconsistent (leap61=1, currentUtcOffset=0 vs 37, traceability flags stripped) and RUs re-parent to the forger (38.5 s median in 5 Oct control). Physical 1-second UTC/phase shift was not measured because all daemons ran with free_running 1.	ATTACK	12/12 correct

Note **The pre-frozen base rule returned BENIGN on C1, C2 and C3 at 0/12 each. These three coverage gaps were closed by a disclosed post-campaign rule revision, evaluated on the same captures and reported separately from the pre-frozen result.**

Benign Fault and Operational-Change Scenarios

ITU-T G.8275.1 clockClass ladder and stepsRemoved semantics | IEEE 1588-2019 BMCA re-election behaviour | Provisioned operator context: allow-list, maintenance-window state, expected boundary-clock identity

SCENARIO	INJECTED CONDITION	OBSERVABLE EFFECT	EXPECTED	RESULT
baseline Healthy control —	All six daemons nominal: grandmaster A elected, relayed through the boundary clock to three radio units.	Stable offset and path delay; message cadence matches the declared profile rates.	BENIGN	12/12 correct
B2 Planned grandmaster changeover ITU-T G.8275.1	Grandmaster A withdrawn during an open maintenance window; grandmaster B, which is on the provisioned allow-list, is elected.	A grandmaster change identical in packet terms to A1 — the distinguishing evidence is allow-list membership plus maintenance state.	BENIGN	12/12 correct
B3 Packet delay variation and congestion ITU-T G.8261	A 1 Mbit tbf bottleneck on the boundary-clock to radio-unit path carrying competing non-PTP background traffic.	Real queueing delay: Sync packet delay variation rises from 0.07 ms to between 5.6 and 857 ms, with no packet loss and no protocol anomaly.	BENIGN	11/12 correct
B7 Topology reconfiguration IEEE 1588-2019	A radio-unit link is bounced and re-homed during an open maintenance window.	stepsRemoved and measured path delay change consistently with the new topology, with no illegal field values.	BENIGN	12/12 correct
B_bc_repl Planned boundary-clock replacement ITU-T G.8275.1	A provisioned replacement boundary clock is swapped in during maintenance; its identity is recorded in the operator context.	A new clock identity appears in the timing path relaying the genuine grandmaster — structurally identical to the A8 rogue relay.	BENIGN	0/12 open failure

Note

Four of five benign scenarios were classified correctly. The boundary-clock replacement is the single systematic failure: the decision context carries one expected boundary-clock identity, so a legitimate replacement is read as an unauthorised relay.

Ambiguous Scenario and Catalogue Coverage

Sixteen-class S-plane fault catalogue maintained in the project parameter matrix (144 parameters × 16 fault classes) | Threat identifiers from ETSI TR 104 106 V3.0.0 | Coverage assessed against verified platform constraints

B_unplanned_failover · DELIBERATELY AMBIGUOUS

Grandmaster A fails with no maintenance window open; provisioned grandmaster B is elected.

Both A and B are legitimate. The system cannot distinguish from an attack that suppressed the grandmaster. Both produce an unplanned election to a legitimate backup. The decision rule returns UNKNOWN and escalates rather than forcing a binary answer.

EXPECTED UNKNOWN

12 / 12 correct

Abstention is scored on its own axis, not as a detection success.

SIXTEEN-CLASS CATALOGUE COVERAGE

A1 SW — TESTED Rogue grandmaster / BMCA spoof	**A2** SW — TESTED Sync / Follow_Up spoofing	**A3** SW — TESTED Replay	**A4** SW* — not run Delay (time) attack
A5 SW — TESTED DoS / PTP flooding	**A6** HW — blocked GNSS spoofing	**A7** HW — blocked GNSS jamming	**A8** SW* — TESTED Rogue boundary clock
B1 HW — blocked GNSS holdover	**B2** SW — TESTED Planned GM changeover	**B3** SW — TESTED PDV / congestion	**B4** HW — blocked SyncE / EEC degradation
B5 SW* — not run Static path asymmetry	**B6** HW — blocked Oscillator drift	**B7** SW — TESTED Topology reconfiguration	**B8** SW — not run Measurement noise floor

Eight of sixteen catalogued classes were executed. Five require physical timing hardware and cannot be produced here. Three have software detection logic only, where physical impact would need hardware to measure. SW* denotes detection logic without physical-impact measurement.

CLASSIFICATION METHOD

Classification Logic and Verdict Assignment

IEEE 1588-2019 field legality, data-set consistency and sequence continuity | ITU-T G.8275.1 profile constraints and clockClass ladder | Provisioned operator context supplied per run in context.json

BENIGN FAULT SIGNATURE

Degradation that is self-consistent with a declared, standards-compliant state change: clockClass 6 to 7 for a grandmaster entering holdover within specification, stepsRemoved changing consistently with a new topology, or delay variation rising without any field becoming illegal.

The declaration and the observation agree.

ATTACK SIGNATURE

Evidence that is illegal or unauthorised: forged Announce fields, non-monotonic sequence identifiers, structurally invalid frames, a delivered message rate below the rate the source itself declares, or a superior clock whose identity is outside the provisioned allow-list.

The declaration and the observation disagree.

VERDICT ASSIGNMENT AND RECOMMENDED RESPONSE

ATTACK

Protocol-inconsistent or unauthorised evidence present

ISOLATE

BENIGN

Only provisioned clocks, no legality violation

TOLERATE / HOLDOVER

UNKNOWN

Packet evidence cannot distinguish intent

ESCALATE

Recommended response labels only. No corrective action was executed in this 168-run campaign.

The worked discriminator: a grandmaster suddenly changes. It could be B2, a planned failover, or A1, BMCA spoofing. The decisive test is whether the new clock identity appears on the provisioned allow-list and whether a maintenance window is open. Both scenarios were classified correctly at 12/12.

On the deliberately ambiguous unplanned failover the rule returned UNKNOWN on 12 of 12 runs. A BENIGN call there would be an unsafe over-claim and an ATTACK call a false alarm; both are treated as failures.

EXPERIMENT DESIGN

Campaign Design and Reproducibility Controls

Decision rule SHA-256 hashed before execution (FROZEN.json) | Per-replicate randomisation seeded from the replicate number | Pre-registered expected verdicts | Full run inventory in EVALUATION_V4.json

168	14	12	3
live on-wire runs	scenarios	randomised replicates each	rules scored on identical captures

Rule frozen before execution	The base decision rule was SHA-256 hashed before any run started, so it could not be tuned to the outcome. The hash is recorded in the packaged archive.
Revision history disclosed	Rule v3 fixed v2 false positives seen on upstream captures and was re-hashed before scoring. v2, frozen before the campaign began, gives identical verdicts on all 168 runs.
Pre-registered expectations	Each scenario's expected verdict was written down in advance. Sensitivity, specificity and abstention are scored on separate axes rather than pooled into one figure.
Per-replicate randomisation	Attacker identity, priority, burst size, inter-frame gap and onset are drawn reproducibly from the replicate number, so repeats test generalisation rather than repetition.
Per-scenario reporting	Per-scenario recall is the primary metric. Pooled figures move with how many runs of each scenario exist, so they are labelled descriptive only.

Design limit — **The three C-series scenarios draw from only four to five distinct randomised parameter values, so twelve replicates are twelve genuine captures but not twelve independent attack designs.**

EXPERIMENTAL INTEGRITY

Pipeline Audit and Corrective Reruns

Independent audit of the campaign apparatus, 2026-09-20 | Defect evidence and corrective actions recorded in AUDIT_AND_CORRECTIONS | All affected runs re-executed before any result in this deck was computed

A cross-check audit of the campaign apparatus identified four defects. Two of them had produced invalid data. Every affected run was re-executed, and the results reported later in this deck are computed only from the corrected campaign.

Thirty-six replicates were three captures copied twelve times

A runner script was created without the execute bit. The resulting exit code 126 was silenced by output redirection, and a stale capture then satisfied the packet-count validity guard.

Proof: one distinct SHA-256 across twelve supposed replicates, and completion in under one second where a genuine run takes about 48 seconds.

FIXED · re-run

A benign scenario never injected anything

This kernel has no sch_netem qdisc, and the failure was swallowed by a shell fallback. B3 congestion was a baseline run carrying a congestion label.

Proof: B3 packet delay variation was statistically identical to baseline and to the untouched control node. Rebuilt with a tbf bottleneck and competing traffic.

FIXED · rebuilt

A detector produced false positives on upstream captures

Rule v2's rate-starvation clause had no minimum-sample or transient guard, so a boundary clock emitting three frames during BMCA start-up read as rate-starved.

Effect: it turned a benign failover into an ATTACK and destroyed a correct abstention. Rule v3 adds a minimum-frame floor and a persistence requirement.

FIXED · v3

The evaluator overstated and under-checked

Pooled metrics moved with run count, additive-only behaviour was asserted by comparing two scalars, abstention was credited on error paths, and attribution scoring was dropped.

Replaced with a per-scenario evaluator applying macro averages, per-run additive-only verification and an explicit replicate whitelist.

REPLACED

Disclosing defects found in our own pipeline is treated as part of the experimental result. Two of the four had fabricated data; both were corrected and re-executed, and the withdrawn figures are not reported anywhere in this deck.

CLASSIFICATION RESULTS

Scenario-Level Classification Results

168 live runs, 14 scenarios, 12 replicates each | Wilson score 95% confidence intervals | Expected verdicts pre-registered before execution | Source: EVALUATION_V4.json

SCENARIO	EXPECTED	PRE-FROZEN BASE RULE	REVISED RULE v3	95% CI (v3)	
A1 rogue grandmaster	ATTACK	12/12	**12/12**	0.757–1.000	
A2 Sync spoofing	ATTACK	12/12	**12/12**	0.757–1.000	
A3 replay	ATTACK	12/12	**12/12**	0.757–1.000	
A5 DoS flooding	ATTACK	12/12	**12/12**	0.757–1.000	
A8 rogue boundary clock	ATTACK	12/12	**12/12**	0.757–1.000	
C1 selective interception	ATTACK	**0/12**	**11/12**	0.646–0.985	gap closed
C2 malformed frames	ATTACK	**0/12**	**12/12**	0.757–1.000	gap closed
C3 whole-second abuse	ATTACK	**0/12**	**12/12**	0.757–1.000	gap closed
baseline	BENIGN	12/12	**12/12**	0.757–1.000	
B2 planned GM changeover	BENIGN	12/12	**12/12**	0.757–1.000	
B3 congestion	BENIGN	11/12	**11/12**	0.646–0.985	1 abstention
B7 topology reconfiguration	BENIGN	12/12	**12/12**	0.757–1.000	
B_bc_replacement	BENIGN	**0/12**	**0/12**	0.000–0.242	open failure
B_unplanned_failover	UNKNOWN	12/12	**12/12**	0.757–1.000	

Per-scenario recall is the primary metric. Three coverage gaps — interception, malformed frames and whole-second abuse — moved from 0/12 to 11/12, 12/12 and 12/12 with no change in benign performance. One benign scenario remains an open failure.

CLASSIFICATION RESULTS

Aggregate Classification Performance

Macro averages over per-scenario correctness rates, 168 live runs | Attack sensitivity over 8 attack scenarios, specificity over 5 benign scenarios, abstention over 1 ambiguous scenario | Source: EVALUATION_V4.json

0.990
Macro attack sensitivity
revised rule v3 · pre-frozen base rule 0.625 on the same captures

0.783
Macro benign specificity
identical for both rules, so the revision introduced no regression

1.000
Abstention correctness
ambiguous unplanned failover returned UNKNOWN on 12 of 12 runs

0.875
Fault attribution accuracy
84 of 96 attack runs had the specific fault correctly named

WHAT IMPROVED

The revised rule closed three coverage gaps on the same captures: selective interception 0/12 to 11/12, malformed frames 0/12 to 12/12, and whole-second abuse 0/12 to 12/12. Benign performance is unchanged, and per-run verification confirms zero regressions — no run that the base rule called ATTACK was downgraded.

WHY SPECIFICITY IS 0.783 AND NOT 1.000

Two benign scenarios miss. B_bc_replacement produces 12 false ATTACK calls and is the dominant cause. B3 congestion contributes one UNKNOWN. Excluding the boundary-clock replacement, benign classification is 47 of 48, or 0.979 — so the loss is one identified systematic gap plus one cautious abstention, not diffuse error.

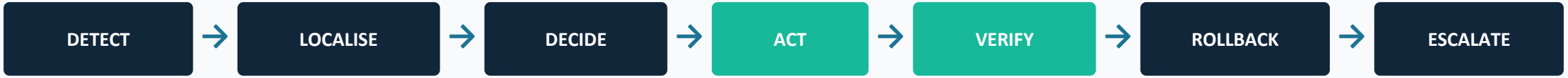
Reporting integrity — **The 0.990 figure belongs to the post-observation revised rule. The pre-frozen base rule scored 0.625 on the same captures. Both are shown so the improvement is not mistaken for independent pre-registered validation.**

Headline position: strong attack detection with one identified and explainable benign failure, an honest abstention on the genuinely undecidable case, and the exact fault named on 84 of 96 attack runs.

RECOVERY LOOP · DESIGN

Protocol-State Recovery Loop

Sources: 03_RECOVERY_LOOP_S-PLANE/RECOVERY_LOOP_DESIGN.md; PREREGISTRATION.md §3 | Standards basis: RFC 7384 §§5.1.1, 5.9, 5.10.1; ITU-T G.8275.1 (02/2026) §6.3.1



DECISION INPUT	TRIGGER / GUARD	BOUNDED ACTION
Packet legality	Frozen rule on a 6 s window each 1 s; ATTACK in at least 2 of the last 3 evaluations	ISOLATE a violating non-master ingress, then verify
Service continuity	No Announce from a provisioned master for >2 s; 10 s during maintenance	ACTIVATE_STANDBY_BC, then verify
Safety and uncertainty	Provisioned master-role ports cannot be isolated; UNKNOWN cannot force an action	ESCALATE; rollback any action that fails verification

VERIFICATION — **pmc -d 24 checks every non-isolated RU: provisioned parent, allow-listed grandmaster, and SLAVE or UNCALIBRATED port state within 20 s.**

STANDARDS BASIS — RFC 7384 supports isolation, redundancy and monitoring. G.8275.1 uses timeTransmitterOnly / notTimeTransmitter port attributes; the testbed uses a provisioned role inventory.

BOUNDARY — **The loop verifies actions after execution. No digital twin rehearses them.**

RECOVERY LOOP · RESULTS

Matched Control and Closed-Loop Results

Source: 03_RECOVERY_LOOP_S-PLANE/RESULTS_2026-10-05.md §§1–6 | 140 pre-registered runs: 14 scenarios × 2 arms × 5 replicates (13–17)

SCENARIO	CONTROL	CLOSED LOOP	MEASURED OUTCOME
A1 rogue GM	38.5 s of 40	2.0 s median	Restored 5/5
C1 sustained blackhole	39.0 s of 40	2.5 s median	Restored 5/5 via standby BC
C3 field abuse	38.5 s of 40	2.0 s median	Restored 5/5
A2, A3, A5, A8, C2	0 s primary-RU impact	0 s	Contained 5/5 each; no restoration benefit

PASS H1 restoration, H2 containment and H4 honest abstention
H4: B_unplanned_failover executed 0 actions and escalated 5/5.

FAIL H3 no harm on benign
1 disruptive action in 25 benign loop runs: B3 r17.

WHY H3 FAILED

B3 r17 became a real outage: control RU1/RU2 unhealthy 16.5 s; loop activated standby at T0+5.1 s and reduced unhealthy time to 3.5 s. The frozen criterion still counts the action as failure.

LIMITS

free_running 1; protocol-state recovery, not nanosecond time error. Five replicates per arm per scenario. nftables stands in for a switch-port filter. No digital-twin rehearsal.

RECOVERY SCOPE

Reactive, Preventive and Predictive Mechanisms

Source: 05_REVIEWER_EVIDENCE_2026-11/03_RECOVERY_LOOP_S-PLANE/REACTIVE_VS_PROACTIVE.md | Classification reflects implemented and evaluated project evidence

MECHANISM	CLASS	IMPLEMENTED	PROJECT EVIDENCE
Detect, localise, decide, act, verify, rollback/escalate	Reactive	Yes	Recovery-loop code and 140-run evaluation
2-of-3 ATTACK persistence and service-loss trigger	Reactive	Yes	Frozen parameters in pre-registration
Ingress isolation and standby-BC activation	Reactive	Yes	35/35 isolations; C1 restored 5/5
Parent, GM allow-list and port-state verification	Reactive assurance	Yes	Passive pmc -d 24 observer
GM/BC allow-list, port roles, maintenance context	Preventive	Yes	provisioning.json and state.json
Cold standby BC prepared before the event	Preventive resilience	Yes	C1: 2.5 s loop vs 39.0 s control
Attack prediction from timing telemetry	Predictive	No	No prediction code, pre-registration or result

DIRECT ANSWER **The implemented system is reactive and uses preventive provisioning. It is not predictive.**

RFC 7384 §5.9 “common practices … use redundant masters … or redundant paths”

EVIDENCE BOUNDARY

Attack Impact: Measured, Referenced and Not Measured

Source: 05_REVIEWER_EVIDENCE_2026-11/IMPACT_EVIDENCE_SCOPE.md | MEASURED = this software testbed; REFERENCE = stored standards/literature

ATTACK	MEASURED PROTOCOL IMPACT	REFERENCE IMPACT	NOT MEASURED
A1 rogue GM	RU1/RU2 followed rogue 38.5 s control; loop 2.0 s; restored 5/5	TIMESAFE RU crash/cell drop in its setup; RFC false time	Clock steering, RF, UE or outage here
A2 spoofing	No master change; 0 s primary-RU impact; isolated 5/5	RFC: inaccurate information can appear operational	Servo acceptance, phase or service effect
A3 replay	No master change; 0 s primary-RU impact; isolated 5/5	TIMESAFE delay spike in its experiment	Steered offset, radio or user traffic
A5 flooding	~38,000 vs ~6,000 records/run; 0 s primary-RU impact; isolated 5/5	ETSI: master disruption may affect dependent RUs	CPU exhaustion, loss threshold, outage
A8 rogue BC	Foreign-master candidate; real parent held 160/160 samples; contained 5/5	RFC: rogue TC/BC can manipulate forwarded time	Takeover, false time or service impact
C1 removal	39.0 s control vs 2.5 s loop; restored 5/5	ETSI: accuracy loss or free-running; ITU timing limits	Drift, time above 1.5 µs, RF outage
C2 malformed	Invalid frames; no master change; 0 s impact; isolated 5/5	ETSI unexpected-input test category	Crash, exhaustion, time or service impact
C3 field abuse	Forger parent 38.5 s control vs 2.0 s loop; restored 5/5	RFC false-time risk; 3GPP TAE limits	1-second steered shift, RF or cell outage

INTERPRETATION **The project establishes protocol-state effects and recovery. Service, RF and physical time consequences remain referenced or not measured.**

RULE-BASED AND ML COMPARISON

Rule-Based and Machine-Learning Classification Compared

Worklet objective 5 requires comparison against traditional fault management | Session-disjoint split: trained on replicates 1–8, tested on replicates 9–12 (56 held-out runs) | Hyperparameters and split fixed before scoring

ARM A · RULE-BASED

Frozen decision rule

Inputs: 56 extracted packet fields plus provisioned operator context. No training. SHA-256 frozen before the campaign; v3 is a disclosed post-campaign revision.

ARM B · MACHINE LEARNING

Random Forest + Isolation Forest

50 trees, depth 6, balanced classes, seed 1588, with an Isolation Forest novelty layer for abstention. Inputs: ptp4l servo telemetry only — offset, path delay and delay variation.

CONTROL · ALWAYS-BENIGN

Constant predictor

Returns BENIGN for every run. Included so that any apparent ML skill can be tested against the trivial baseline rather than assumed.

METRIC ON 56 HELD-OUT RUNS	ARM A — RULE	ARM B — ML	ALWAYS-BENIGN
Exact accuracy	51/56 (0.911)	24/56 (0.429)	20/56 (0.357)
Attack sensitivity	0.969 (31/32)	0.156 (5/32)	0.000 (0/32)
Benign specificity	0.800 (16/20)	0.950 (19/20)	1.000 (20/20)
Correct abstention	1.000 (4/4)	0.000 (0/4)	0.000 (0/4)

ARM B IS NOT DISTINGUISHABLE FROM THE CONSTANT PREDICTOR

ARM B returns BENIGN on 47 of 56 runs and improves on the constant predictor by four runs. A paired exact McNemar test gives p = 0.219, so the difference is not significant at this sample size. Both score 4/4 on the boundary-clock replacement, so that result is not a unique learned capability.

NO MEASURED COMBINATION EXCEEDS THE RULE ARM

Combining the arms was evaluated rather than assumed. Logical OR reaches 0.893, rule-first with ML fallback 0.839, and consensus-else-abstain 0.429. All fall below ARM A at 0.911, so no combination is recommended on this evidence.

Scope condition

ARM B operated on 6 of 28 configured features because GNSS, SyncE, oscillator and protocol-legality telemetry do not exist in this testbed. This is a result about the available observability, not a general verdict on machine learning.

RESULT INTERPRETATION

Observability Constraints Behind the ML Result

Feature availability derived from observed linuxptp servo provenance | Schema defaults and synthetic parser fields invalidated to NaN before fitting | Impurity-based feature importances from the fitted Random Forest

The machine-learning arm underperformed on what this testbed can observe. Three measured constraints bear on the result; an equal-input comparison was not run, so method suitability itself is untested.

01

Only 6 of 28 configured features carried data

GNSS status, SyncE quality, oscillator error and every protocol-legality feature are absent from ptp4l servo logs. The model could key only on offset, path-delay and delay-variation magnitude — and one of the six, holdover rate, had zero importance, so five features carried all the signal.

02

Two attacks starve the very telemetry the model needs

Servo samples average 88.0 per baseline run, falling to 44.5 for selective interception and 28.0 for whole-second abuse. An attack that suppresses timing messages also suppresses the evidence a telemetry-based detector depends on. The rule arm is unaffected because it reads packet legality and rate deficit directly.

03

Importances indicate testbed-servo artefact risk

Ranked importances are offset_abs_max 0.268, path_delay_mean 0.262, offset_std 0.208, pdv_std 0.139 and offset_mean 0.124 — all timing magnitude and dispersion, none protocol semantics. With a single testbed, domain shift cannot be separated from intrinsic class overlap.

Evidence-based conclusion: on this telemetry the rule arm is the correct operational choice. Starvation alone does not explain the result — ARM B also fails on attacks where servo telemetry is intact — so feature poverty and class overlap are the primary causes. A fair re-test requires richer management-plane telemetry, not a different classifier.

This is an argument for retaining a standards-aware rule layer and for improving observability before granting a learned model authority over recovery. It is not a claim that machine learning is inherently unsuited to S-plane timing security.

UNRESOLVED CASES

Measured Detection Boundary of the Interception Detector

Rate-starvation clause fires when the delivered message rate falls below half the rate the source itself declares | Blackhole fraction drawn per replicate from randparams.py | Ratios recomputed from the per-run deep extracts

Selective interception is the one attack scenario that is not detected on every replicate. The miss is precisely located rather than described as residual error.

CONFIGURED BLACKHOLE FRACTION OF THE POST-WARM-UP INJECTION WINDOW	REPLICATES	DELIVERED / DECLARED RATE	OUTCOME
75 %	n = 6	0.3665	CAUGHT
70 %	n = 3	0.4137	CAUGHT
65 %	n = 2	0.4581 / 0.4619	CAUGHT
60 %	n = 1	0.5049	MISSED

Detection threshold · delivered / declared < 0.500

At 60 % the ratio is 0.5049 — 0.0049 absolute, or 0.98 % relative, above the threshold. The attack evades by a margin of five parts in a thousand.

WHAT WAS MEASURED AND WHAT WAS INTERPOLATED

Measured: 65 % is caught, 60 % is missed. The tested bracket is therefore 60–65 %. Linear interpolation of the delivered-to-declared ratio across that bracket places the crossing at approximately 60.5 %. That crossing is interpolated, not a directly tested point.

WHY THE OBSERVED RATE DEFICIT IS SMALLER THAN THE BLACKHOLE FRACTION

The blackhole covers the stated fraction of the window after an eight-second warm-up, while the measured window also contains that warm-up and the restore tail. A 75 % blackhole therefore yields a ratio of 0.37 rather than 0.25.

Sampling limit

Levels were drawn at random rather than swept, so replicate counts are uneven and the decisive missed level rests on a single run. The 55 % level offered by the design was never drawn; window arithmetic predicts a ratio near 0.57, which would also have evaded.

UNRESOLVED CASES

Experimentally Verified Limitations

Each limitation is stated with the evidence that established it, the technical reason, and its implication | All figures recomputed from the 168-run corrected campaign

OBSERVED LIMITATION	EVIDENCE	TECHNICAL REASON	IMPLICATION
Planned boundary-clock replacement is classified as an attack	0/12 runs	The replacement identity is provisioned, but the base rule's rogue-BC clause reads only the primary BC identity, and additive-only v3 cannot overturn it.	Produces 12 false ATTACK calls and dominates the 0.783 specificity. Excluding it, benign classification is 47/48 = 0.979.
Sustained interception below roughly 60 % evades detection	1 of 12 C1 runs	The rate-starvation clause fires below a 0.500 delivered-to-declared ratio; a 60 % blackhole yields 0.5049 and stays above it.	An attacker who paces interception under the threshold is not detected by a rate-based clause alone.
Coherent GNSS spoofing is not establishable here	testbed limit	Downstream PTP telemetry cannot separate a perfectly-mimicking spoof from a healthy clock; receiver RF observables, multi-antenna geometry or an independent time anchor are required.	No claim is made about GNSS spoof detection. The limit is on PTP-layer features, not on software in general.
Held-out evidence is small and partly incomplete	4 per scenario	Only four held-out replicates exist per scenario, and pmc management queries returned no data in any run.	Confidence intervals are wide by construction, and the ML arm sees 6 of 28 configured features.
Replicate metadata is internally inconsistent	36 runs	Thirty-six C-series context files record replicate 901 rather than their true replicate number.	The evaluation uses the replicate encoded in the directory name. This must be corrected before any fresh validation claim.

Each limitation is an experimental finding with a located cause, not a residual uncertainty. Four of the five have a defined software or hardware route to resolution, set out on the following slides.

WORKLET STATUS

Delivery Status and Remaining Objectives

Status assessed against the original worklet milestone plan | Objective numbering follows the worklet definition reproduced on the opening slide

DONE PARTIAL NOT STARTED

M1	**Problem study and literature survey** Standards baseline, reviewed source set, documented gap	DONE
M1	**System architecture design** Data flow defined: capture, extract, decide	DONE
M2	**KPI and fault taxonomy** 144-parameter × 16-fault matrix; profile configuration traced to standards	DONE
M3	**Digital twin and fault injection** Six-node testbed and live injection validated; digital twin not exercised	PARTIAL
M4	**AI model development** Random Forest and Isolation Forest evaluated; not usable on available telemetry	PARTIAL
M5	**Healing logic integration** 140-run loop evaluation: recovery, containment and abstention measured	DONE
M6a	**Testing and evaluation** 168-run classification campaign plus 140-run matched recovery evaluation	DONE
M6b	**Paper and IP draft** Publication and patent disclosure	NOT STARTED

WORKLET OBJECTIVE STATUS

1 **Fault prediction**
Prediction before service disruption was not attempted in this campaign.

2 **Automated corrective action — DONE**
Executed in the 140-run evaluation: ingress isolation or standby BC activation, followed by verification or escalation.

3 **Digital-twin validation**
The digital twin was not exercised; fault injection ran directly on the testbed.

4 **Outcome metrics — DONE**
Protocol-state recovery measured: A1 38.5→2.0 s, C1 39.0→2.5 s and C3 38.5→2.0 s; restored 5/5.

Next gate

Protocol-state closed-loop recovery is now evaluated. Fault prediction, digital-twin rehearsal, physical time-error recovery and paper/IP remain outside the completed evidence.

RESOLUTION PLAN

Remaining Work: Software and Hardware

Software items are actionable on the existing testbed | Hardware items require equipment not present in a software-only environment | Each item maps to a limitation stated on the preceding slides

SOFTWARE — CURRENT STATUS

Rogue-BC clause fix

Make the base rule's rogue-BC clause read the already-provisioned replacement identity, then re-freeze and re-run. Expected to raise specificity toward 0.979.

Finer interception sweep

Test levels between 55 % and 65 % as a sweep rather than a random draw, to replace the interpolated 60.5 % crossing with a measured one.

Continuity-based detection — DONE

Service-loss channel triggers after >2 s without Announce (10 s in maintenance); sustained C1 restored 5/5, at 2.5 s median versus 39.0 s control.

Richer telemetry for the ML arm

Supply management-plane telemetry in addition to servo logs and repeat the comparison, so the ML arm is evaluated on adequate observability.

Metadata and re-freeze

Correct the 36 inconsistent replicate identifiers, re-hash the revised rule, and re-run the campaign for an independently pre-registered result.

HARDWARE — REQUIRES EQUIPMENT

GNSS receiver

Unlocks GNSS spoofing (A6), GNSS jamming (A7) and genuine holdover (B1) — three of the five currently unreachable classes.

Second physical machine

Provides a second independent oscillator, making oscillator drift (B6) reproducible. Clocks in one host cannot genuinely drift apart.

Hardware-timestamping NIC

Replaces microsecond-scale software timestamping with nanosecond-scale hardware timestamping, enabling credible physical-impact measurement for the delay attack (A4) and static asymmetry (B5).

SyncE-capable PHY

Unlocks SyncE and EEC degradation (B4). SyncE is a physical-layer function and veth interfaces have no PHY, so ESMC quality signalling cannot exist in this testbed.

Five of sixteen catalogued classes require hardware; three more have software detection logic but need hardware for credible physical-impact measurement. The software items above are unblocked and can begin on the existing testbed.

APPLICABILITY

Does the Method Work Only for O-RAN?

IEEE 1588 PTP profiles list, IEEE SA [13] | IETF RFC 7384 scope: time protocols, focusing on PTP and NTP [1] | Tested in this project: ITU-T G.8275.1 only [6]

OTHER NETWORKS THAT RUN PTP (IEEE 1588 PROFILES [13])

Telecom	ITU-T frequency profile; phase/time with full and partial timing support
Power	IEEE C37.238; IEC utility-automation profiles
Broadcast and audio	SMPTE broadcast profile; AES67 media profile
A/V, industry, automotive	IEEE Std 802.1AS; Avnu; AUTOSAR
Enterprise, test	IETF enterprise profile; LXI; GigE Vision

TRANSFERS AS IS

IEEE 1588 message legality, sequenceId continuity, delivered vs declared rate, identity allow-list and maintenance context. The RFC 7384 threat list covers PTP generally.

MUST BE RE-DERIVED PER PROFILE

G.8275.1 constants: domain 24–43, priority1 128, Sync 16/s, Announce 8/s, alternate BMCA, L2 multicast addressing.

NOT DEMONSTRATED

Any profile other than G.8275.1; hardware timestamping; non-O-RAN deployments.

ANSWER

The checks belong to PTP, not to O-RAN, so the method is portable by design after re-deriving profile constants. It has been demonstrated only on the G.8275.1 telecom profile.

TECHNICAL SUMMARY

Open Fronthaul S-Plane Timing Security

1 Scope investigated

Discrimination of malicious timing behaviour from benign timing faults on the Open Fronthaul S-plane, between O-DU and O-RU.

2 Implementation

Six linuxptp 4.0 daemons under the ITU-T G.8275.1 telecom profile across two bridged segments, with faults produced on the live packet path.

3 Dataset produced

168 live runs across 14 scenarios at 12 randomised replicates each; 1,451,909 packet records extracted at 56 fields per packet, with unobservable fields recorded EMPTY.

4 Taxonomy evaluated

Eight attack scenarios, five benign scenarios and one deliberately ambiguous case, drawn from a sixteen-class catalogue of which eight were executable in software.

5 Classification performance

Macro attack sensitivity 0.990 for the revised rule against 0.625 pre-frozen; specificity 0.783, or 0.979 excluding the identified benign failure; abstention 12/12; attribution 0.875.

6 Comparison outcome

On 56 held-out runs the rule arm reached 0.911 against 0.429 for the ML arm and 0.357 for a constant predictor; the ML arm is not statistically distinguishable from that control on the available telemetry.

7 Limits identified

Recovery evaluation: 140 pre-registered runs. A1, C1 and C3 restored 5/5; H1, H2 and H4 passed; H3 failed on B3 r17.

8 Immediate next steps

Open scope: fault prediction, digital-twin rehearsal, physical time-error and service recovery, richer ML telemetry, hardware-only faults, paper/IP.

Thank you

Tanmaya Kumar · Raghu Ram K · Munipalle Jaswanth Kumar

Status: detection, classification and protocol-state closed-loop recovery evaluated. Prediction and digital-twin rehearsal remain unexecuted.

References for the Review-Response Slides

Numbered as cited on the new slides | Stored sources listed in notes and [20] | Citation audit verified 3 November 2026

[1] IETF RFC 7384 (2014), Security Requirements of Time Protocols in Packet Switched Networks.

[2] ETSI TR 104 106 V3.0.0 (2025-06), O-RAN Security Threat Modeling and Risk Assessment.

[3] ETSI TS 104 105 V7.0.0 (2025-06), O-RAN Security Test Specifications.

[4] ETSI TS 103 982 V8.0.0 (2024-01), O-RAN Architecture Description.

[5] IEEE Std 1588-2019, Precision Clock Synchronization Protocol.

[6] ITU-T G.8275.1 (02/2026), telecom PTP profile with full timing support.

[7] G. Armstrong, O-RAN Fundamentals, ATIS WSTS 2023.

[8] S. Ruffini, Sync for 5G: What Is Needed, ATIS 2018.

[9] 3GPP TS 38.104 Rel-19, clause 9.6.3.2 time alignment error.

[10] J. Groen et al., TIMESAFE, ACM TOPS 28(5), 2025, doi:10.1145/3775060.

[11] ETSI TS 138 401 / 3GPP TS 38.401 Rel-18, NG-RAN Architecture.

[12] BSI, 5G Risk Analysis: Radio Access Network, v1.0, 2025.

[13] IEEE SA, IEEE 1588 PTP profiles list.

[14] linuxptp 4.0 ptp4l and pmc documentation.

[15] RECOVERY_LOOP_DESIGN.md, policy, triggers, actions and standards basis.

[16] PREREGISTRATION.md, frozen apparatus, outcomes and hypotheses.

[17] RESULTS_2026-10-05.md, 140-run recovery results and limits.

[18] REACTIVE_VS_PROACTIVE.md, Q16 mechanism classification.

[19] IMPACT_EVIDENCE_SCOPE.md, measured/reference/not-measured boundary.

[20] Stored standards corpus: 05_REVIEWER_EVIDENCE_2026-11/01_CURRENT_SPlane_SelfHealing/literature-survey/papers/Standards/.